DRUGS · LOCAL ANAESTHETICS AND VASOCONSTRICTORS

Adrenaline

Adrenaline (epinephrine) appears in ENT in three forms. As a vasoconstrictor, it is mixed with local anaesthetic for infiltration and soaked onto pledgets to dry the surgical field. Nebulised, it shrinks oedematous airway mucosa in croup and post-extubation stridor. Injected intramuscularly, it is the life-saving first treatment for anaphylaxis. The concentrations differ by a factor of a thousand, and confusion between them causes harm.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

ADRENALINE

	SUMMARY
Class	Catecholamine: alpha and beta adrenoceptor agonist
Concentrations	1:1,000 = 1 mg/ml (intramuscular anaphylaxis, nebulised, topical pledgets); 1:10,000 = 100 micrograms/ml (intravenous in cardiac arrest, some topical use); 1:80,000 to 1:200,000 = 12.5 to 5 micrograms/ml (with local anaesthetic infiltration)
ENT uses	Haemostasis in surgery (infiltration and pledgets); epistaxis (pledgets); croup and post-extubation stridor (nebulised); anaphylaxis and allergic angioedema (intramuscular)

	SUMMARY
Anaphylaxis dose (intramuscular, 1:1,000, anterolateral thigh)	Adult and child over 12: 500 micrograms (0.5 ml); 6 to 12 years: 300 micrograms (0.3 ml); 6 months to 6 years: 150 micrograms (0.15 ml); under 6 months: 100 to 150 micrograms; repeat after 5 minutes if no better
Croup (nebulised)	400 micrograms/kg of 1:1,000 (maximum 5 mg), repeated after 30 minutes if needed
Topical	Pledgets soaked in 1:1,000 (or 1:10,000) in endoscopic sinus surgery

CHAPTER TWO

2 Using it in ENT

Practice.

2.1 Surgical haemostasis

Infiltrated with lidocaine, adrenaline constricts arterioles; wait 7 to 10 minutes. Cotton pledgets with 1:1,000 adrenaline in endoscopic sinus surgery reduce bleeding and improve the view; count and remove them. Absorption from the nasal mucosa can cause tachycardia and hypertension, especially with high-concentration solutions.

2.2 Croup and stridor

Nebulised adrenaline reduces mucosal oedema within minutes in **severe croup** or post-extubation stridor; it wears off in about 2 hours, so observe the child for at least 2 to 4 hours afterwards; dexamethasone treats the underlying swelling.

2.3 Anaphylaxis

Recognise airway, breathing or circulation problems with skin or mucosal changes after exposure to a trigger (drugs, latex, chlorhexidine in theatre). Give **intramuscular adrenaline immediately**, lie the patient flat (or sit if breathing is difficult), give oxygen and fluids; repeat after 5 minutes; start an adrenaline infusion for refractory anaphylaxis under critical care. Antihistamines and steroids are not first-line. Take mast-cell tryptase samples and refer to allergy services.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Tachycardia, arrhythmia, hypertension, anxiety, tremor and pallor; myocardial ischaemia in patients with coronary disease; tissue necrosis if injected into compromised tissue or flaps. **Never give 1:1,000 adrenaline intravenously** in anaphylaxis outside critical care.

3.2 Interactions

Non-selective beta blockers (severe hypertension with reflex bradycardia, and poor response in anaphylaxis: glucagon may help), tricyclics and MAOIs (enhanced pressor response), halothane (arrhythmias).

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

> WHAT THE EXAMINER IS LOOKING FOR

- The **concentrations** and what each is used for.
- **Anaphylaxis** doses by age and route.
- Nebulised adrenaline in croup: dose, duration and observation.

4.2 Common pitfalls

COMMON PITFALLS

- Intravenous instead of intramuscular adrenaline in anaphylaxis.
- Delaying adrenaline to give antihistamines.
- Discharging a child soon after nebulised adrenaline.

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